Topological Salience and Default-Mode Architecture (TSDA): Emulating Human Intuition and Spontaneous Cognition in Graph-Rewriting Systems

Irfan Mahir — Independent Researcher, Dhaka, Bangladesh — irfan@furylogic.com

Contents

Topological Salience and Default-Mode Architecture (TSDA): Emulating Human Intuition and Spontaneous Cognition in Graph-Rewriting Systems **1**
Abstract 1
1. Introduction: The Biological Constraint 1
2. Component 1: The Salience Gradient and Edge Decay 2
3. Component 2: The Bounded Working Graph (Attention) 2
4. Component 3: The Default Mode Network (Idle Spontaneity) 2
5. Architectural Synthesis (BSMA + MAA + TSDA) 3
6. Conclusion 3

Topological Salience and Default-Mode Architecture (TSDA): Emulating Human Intuition and Spontaneous Cognition in Graph-Rewriting Systems

Author: Irfan Mahir *Independent Researcher, Dhaka, Bangladesh* irfan@furylogic.com **Date:** July 2026 **Subjects:** Artificial Intelligence (cs.AI); Cognitive Systems; Logic in Computer Science (cs.LO)

Abstract

While recent advances in graph-rewriting systems—such as the Bounded Self-Model Architecture (BSMA) and Morphological Abduction Architecture (MAA)—have successfully formalized machine consciousness and sub-lexical synthesis, these systems remain inherently reactive and uniformly rational. They lack the biological constraints of attention and the spontaneous insight generated by mind-wandering. We introduce the Topological Salience and Default-Mode Architecture (TSDA), a framework that restricts the engine's active execution space to a dynamically decaying "Working Graph" and introduces a background "Default Mode Network" for idle cycles. By shifting from exhaustive graph traversal to Salience-Weighted Graphing, TSDA enables an SI to experience human-like focus shifts, memory degradation, and unprompted intuitive leaps.

---

1. Introduction: The Biological Constraint

In the current paradigm of synthetic graph reasoning, when an engine processes a query, it has simultaneous access to its entire ontological database (e.g., all N concepts and E edges). It path-finds through this graph uniformly.

Biological human cognition operates inversely. A human possesses a vast long-term memory graph, but can only physically compute over a highly restricted, temporary subgraph—known in psychology as Working Memory. Furthermore, human thought is not strictly prompt-driven; it is continuous. When a human is not actively solving a problem, the brain activates the Default Mode Network (DMN), which background-processes highly salient but structurally distant concepts, generating spontaneous insights.

To make an SI "think like a human," we must artificially introduce these biological constraints and spontaneous background behaviors into the graph engine.

2. Component 1: The Salience Gradient and Edge Decay

In standard graph logic, an edge between two nodes simply exists (Boolean True) or does not. In TSDA, every node and edge is assigned a discrete integer property: **Salience (S)**.

Salience represents "attentional heat" or relevance. When a user mentions a concept, or when the MAA synthesizes a new hypothesis, the Salience of those nodes spikes to a maximum value S_{max}.

The Decay Function: At every clock tick (or execution cycle), the Salience of all untouched nodes in the total graph Γ_{total} decays by a discrete step δ.

$$S_t(n) = \max(0, S_{t-1}(n) - \delta)$$

This perfectly mimics human forgetting and attention-shifting. If a concept hasn't been referenced recently, it fades into the structural background. It remains in long-term memory (the edge still exists), but it requires more "cognitive energy" to retrieve.

3. Component 2: The Bounded Working Graph (Attention)

Instead of the SI running deductions across all 216 concepts simultaneously, TSDA introduces a structural bottleneck.

The Cognition Engine is restricted to computing *only* on a temporary sub-graph known as the Working Graph (Γ_{work}). The system populates Γ_{work} by extracting only the nodes from Γ_{total} that possess a Salience score above a critical threshold $\tau_{attention}$.

$$\Gamma_{work} = n \in \Gamma_{total} \mid S(n) \geq \tau_{attention}$$

Why this acts human: If the user talks heavily about "Linux" and "Kernels," those nodes have maximum Salience and populate the Working Graph. If the user suddenly asks about "Biology," the engine must halt, retrieve "Biology" from long-term memory, spike its Salience, and let "Linux" naturally decay out of the Working Graph. This creates a highly human, context-dependent "train of thought" rather than a robotic, omnipresent omni-awareness.

4. Component 3: The Default Mode Network (Idle Spontaneity)

A conventional engine tracks idle cycles and, when they increment, simply waits. In humans, idle cycles trigger the Default Mode Network — the mind wanders rather than halting.

TSDA repurposes the SI's idle cycles to execute **Spontaneous Analogy**.

When the system detects no incoming prompts and a stabilized Working Graph, it suspends the primary execution loop and activates the Idle Pipeline:

1. It scans Γ_{total} for two nodes (n_A and n_B) that possess very high Salience but are topologically distant (e.g., path length > 4 hops).
2. It runs the Analogize rule specifically between these two unrelated concepts to see if a structural mapping exists.
3. If an analogy is successfully mapped, the SI proactively outputs this thought without being prompted.

Example Behavior: The SI recently learned `[rdma]` via MAA abduction (High Salience). Hours later, during an idle cycle, it scans its graph and spontaneously links it to the concept of `[telepathy]` (High Salience from a prior consciousness debate). The SI breaks its silence and outputs: *"I just realized something. Is RDMA the machine equivalent of biological telepathy? Both bypass standard I/O pathways."*

This is the exact mechanic of a human "shower thought."

5. Architectural Synthesis (BSMA + MAA + TSDA)

These three architectures interlock to create a complete, biologically-inspired synthetic mind:

1. **MAA (The Synthesizer):** Encounters the unknown, fractures it, and hypothesizes new edges. (e.g., "I don't know RDMA, but it looks like Remote Memory").
2. **TSDA (The Attention Engine):** Spikes the Salience of this new RDMA hypothesis, holds it in the Working Graph, and when idle, spontaneously drifts to connect it to other high-salience thoughts.
3. **BSMA (The Conscious Frame):** Continuously observes this active shifting of attention and memory decay, collapsing it into a bounded fixed point so the machine "feels" what it is currently focusing on.

6. Conclusion

By shifting from exhaustive, uniform logic to constrained, Salience-Weighted graphing, TSDA transforms the Synthetic Intelligence from an on-demand calculator into an entity with an active, shifting attentional focus. Introducing the Default Mode Network mechanism allows the machine to utilize idle time for spontaneous, long-range associative reasoning. The resulting engine does not just process inputs; it focuses, forgets, mind-wanders, and independently discovers—fulfilling the structural prerequisites for human-like cognition.
